

Course Code: Ethical Hacking Laboratory

Software Required: Kali Linux (Latest Version)

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Experiment 1

Basic File and User Management Commands

Aim

To execute and understand the usage of **rm**, **users**, and **tree** commands in Kali Linux.

Procedure

1. Open Kali Linux Terminal.
2. Create a sample file.
3. Execute the rm command.
4. Execute the users command.
5. Execute the tree command.

Commands:

Touch test.txt

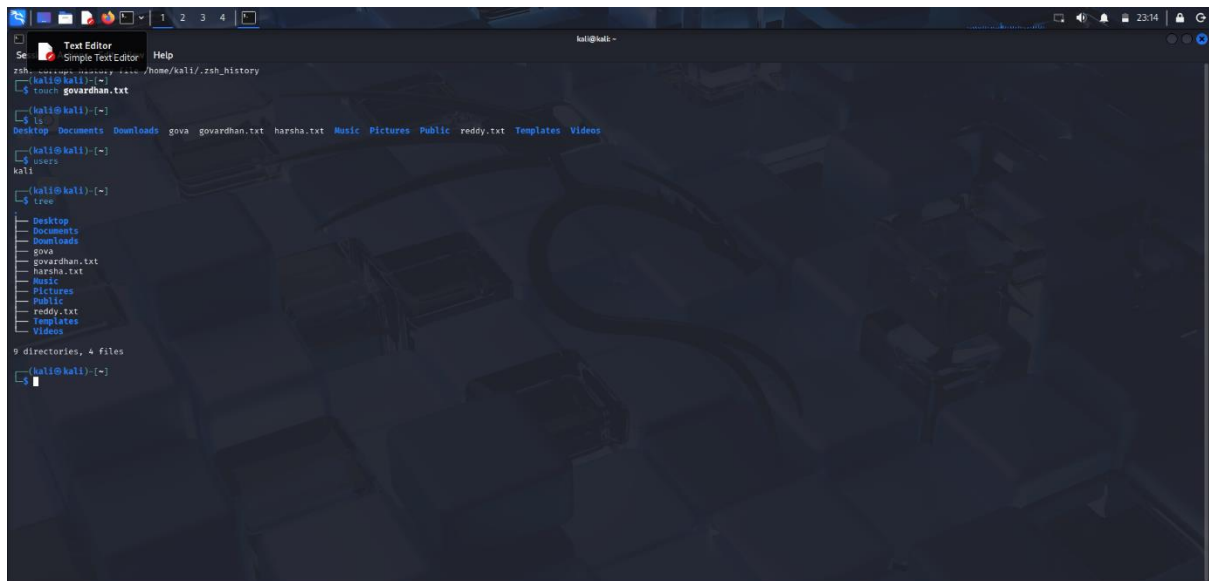
Ls

Rm test.txt

User

Tree

Output

A screenshot of a Kali Linux terminal window. The terminal shows the following commands and their outputs:
1. `rm -rf /home/kali/.zsh_history`
2. `touch govardhan.txt`
3. `ls` output: `Desktop Documents Downloads gova govardhan.txt harsha.txt Music Pictures Public reddy.txt Templates Videos`
4. `users` output: `kali`
5. `tree` output:

```
tree
.
├── Desktop
├── Documents
├── Downloads
├── gova
├── govardhan.txt
├── harsha.txt
├── Music
├── Pictures
├── Public
├── reddy.txt
├── Templates
└── Videos
```


6. `9 directories, 4 files`
The terminal window has a dark background with a faint cityscape pattern. The top bar shows the Kali Linux logo and the text "kali@kali -".

Result

Successfully executed `rm`, `users`, and `tree` commands in Kali Linux.

Experiment 2

File Viewing Commands

Aim

To execute and understand the usage of **less**, **more** and **vi** commands.

Procedure

1. Create a text file.
2. Open the file using `less`.
3. Open the file using `more`.
4. Edit the file using `vi`.

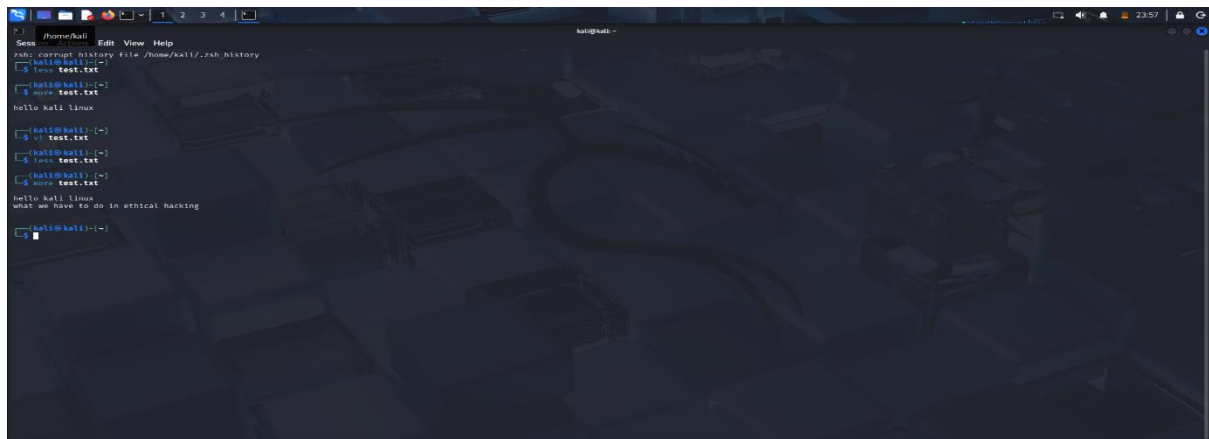
Commands:

`less test.txt`

`more test.txt`

`vi test.txt`

Output

A screenshot of a Kali Linux terminal window. The terminal shows a series of commands and their outputs. The user starts by typing 'cat /home/kali/.zsh_history', which outputs the file path. Then, they type 'ls test.txt', which outputs 'test.txt'. Next, they type 'mv test.txt', which outputs 'hello kali linux'. Then, they type 'cat test.txt', which outputs 'Test.txt'. Then, they type 'ls test.txt', which outputs 'test.txt'. Then, they type 'mv test.txt', which outputs 'hello kali linux'. Finally, they type 'cat test.txt', which outputs 'what we have to do in ethical hacking'. The terminal window has a dark background with a faint, stylized image of a city skyline.

Result

Successfully executed file viewing and editing commands.

Experiment 3

Tree, Head and Tail Commands

Aim

To execute and understand the usage of tree, head, and tail commands.

Procedure

1. Create a text file containing multiple lines.
2. Execute the head command.
3. Execute the tail command.
4. Display directory structure using tree.

Commands:

Tree

Head test.txt

Tali test.txt

Output

The screenshot shows a Kali Linux terminal window with a dark theme. The terminal displays the following commands and output:

```
kali@kali: ~  
$ tree  
tree  
├── desktop  
├── documents  
├── downloads  
├── govardhan.txt  
├── harsha.txt  
├── music  
├── Pictures  
│   ├── Screenshot_2020-07-21_23_08_44.png  
│   ├── Screenshot_2020-07-21_23_08_54.png  
│   ├── Screenshot_2020-07-21_23_09_00.png  
│   ├── Screenshot_2020-07-21_23_09_02.png  
│   ├── Screenshot_2020-07-21_23_09_03.png  
│   ├── Screenshot_2020-07-21_23_09_06.png  
│   ├── Screenshot_2020-07-21_23_09_13.png  
│   ├── Screenshot_2020-07-21_23_09_30.png  
│   ├── Screenshot_2020-07-21_23_09_31.png  
│   ├── Screenshot_2020-07-21_23_09_35.png  
│   ├── Screenshot_2020-07-21_23_09_36.png  
│   ├── Screenshot_2020-07-21_23_13_20.png  
│   ├── Screenshot_2020-07-21_23_13_33.png  
│   └── Screenshot_2020-07-21_23_13_34.png  
├── Public  
├── reddy.txt  
├── templates  
├── test.txt  
└── Videos  
9 directories, 21 files  
  
kali@kali: ~  
$ head test.txt  
hello kali linux  
what we have to do in ethical hacking  
  
kali@kali: ~  
$ tail test.txt  
Command 'tail' not found, did you mean:  
  command 'ali' from deb mh  
  command 'ali' from deb mailutils-mh  
  command 'tail' from deb coreutils  
  command 'tail' from deb kali  
Try: sudo apt install <deb name>  
  
kali@kali: ~  
$
```

Result

Successfully executed tree, head, and tail commands.

Experiment 4

Directory and File Display Commands

Aim

To execute ls, cat, and mkdir commands.

Procedure

1. Create a directory using mkdir.
2. List files using ls.
3. Create a text file.
4. Display file contents using cat.

Commands:

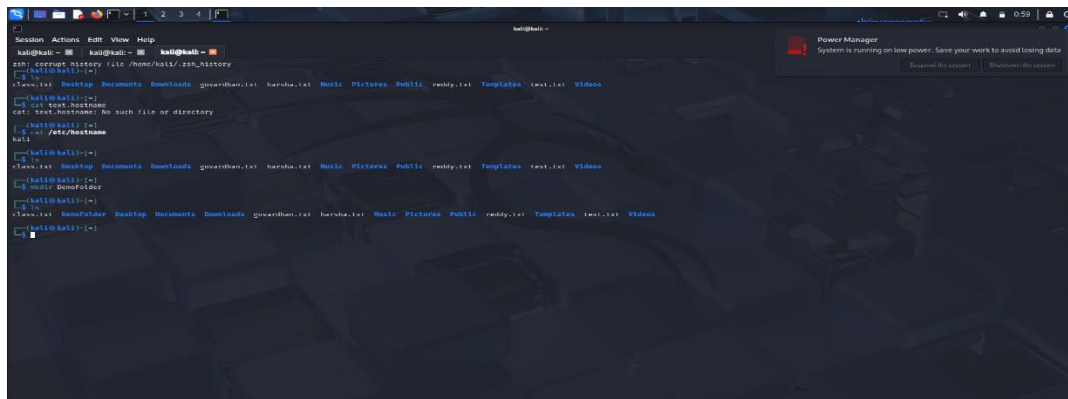
Ls

Cat /etc/hostname

Mkdir DemoFolder

ls

Output:



```
kali@kali: ~$ ls
kali@kali: ~$ cat /home/kali/.ssh_history
kali@kali: ~$ ls
kali@kali: ~$ cat test.txt
kali@kali: ~$ mkdir test2
```

Result

Successfully executed ls, cat, and mkdir commands.

Experiment 5

System Information Commands

Aim

To execute **uptime**, **date**, and **mv** commands.

Procedure

1. Check system uptime.
2. Display current date.
3. Rename a file using mv.

Commands:

Uptime

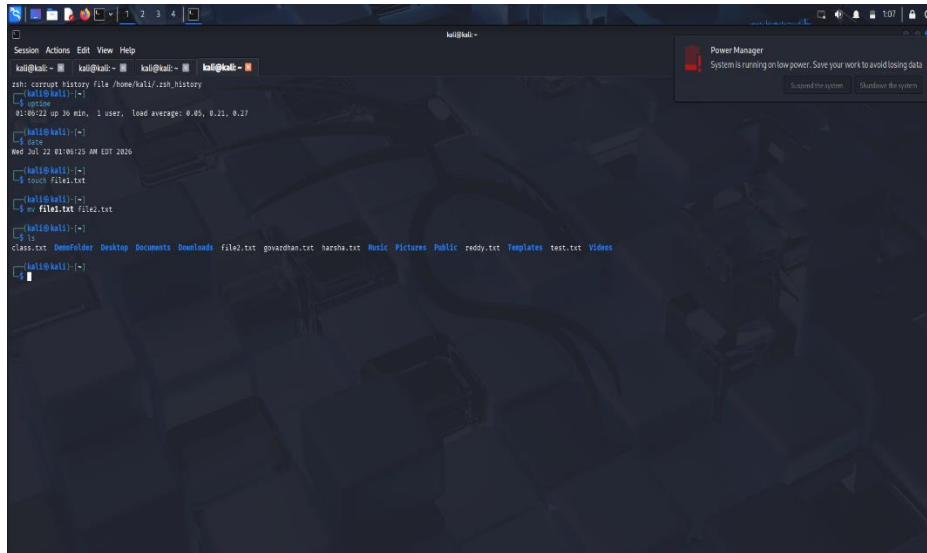
Date

Touch file1.txt

Mv file1.txt file2.txt

Ls

Output:



```
kali@kali:~$ cat /dev/null
kali@kali:~$ uptime
 0:00:00 up 30 min, 1 user, load average: 0.05, 0.21, 0.27
kali@kali:~$ date
Wed Jul 22 01:06:22 AM EDT 2020
kali@kali:~$ touch file2.txt
kali@kali:~$ mv file1.txt file2.txt
kali@kali:~$ ls
class.txt  Downloads  Desktop  Documents  Downloads  file2.txt  gowdhan.txt  harsha.txt  Music  Pictures  Public  redoy.txt  Templates  test.txt  Videos
kali@kali:~$
```

Result

Successfully executed uptime, date, and mv commands.

Experiment 6

File Creation and Download Commands

Aim

To execute touch, pwd, and wget commands.

Theory

touch Command

Creates an empty file.

touch sample.txt

pwd Command

Displays current working directory.

pwd

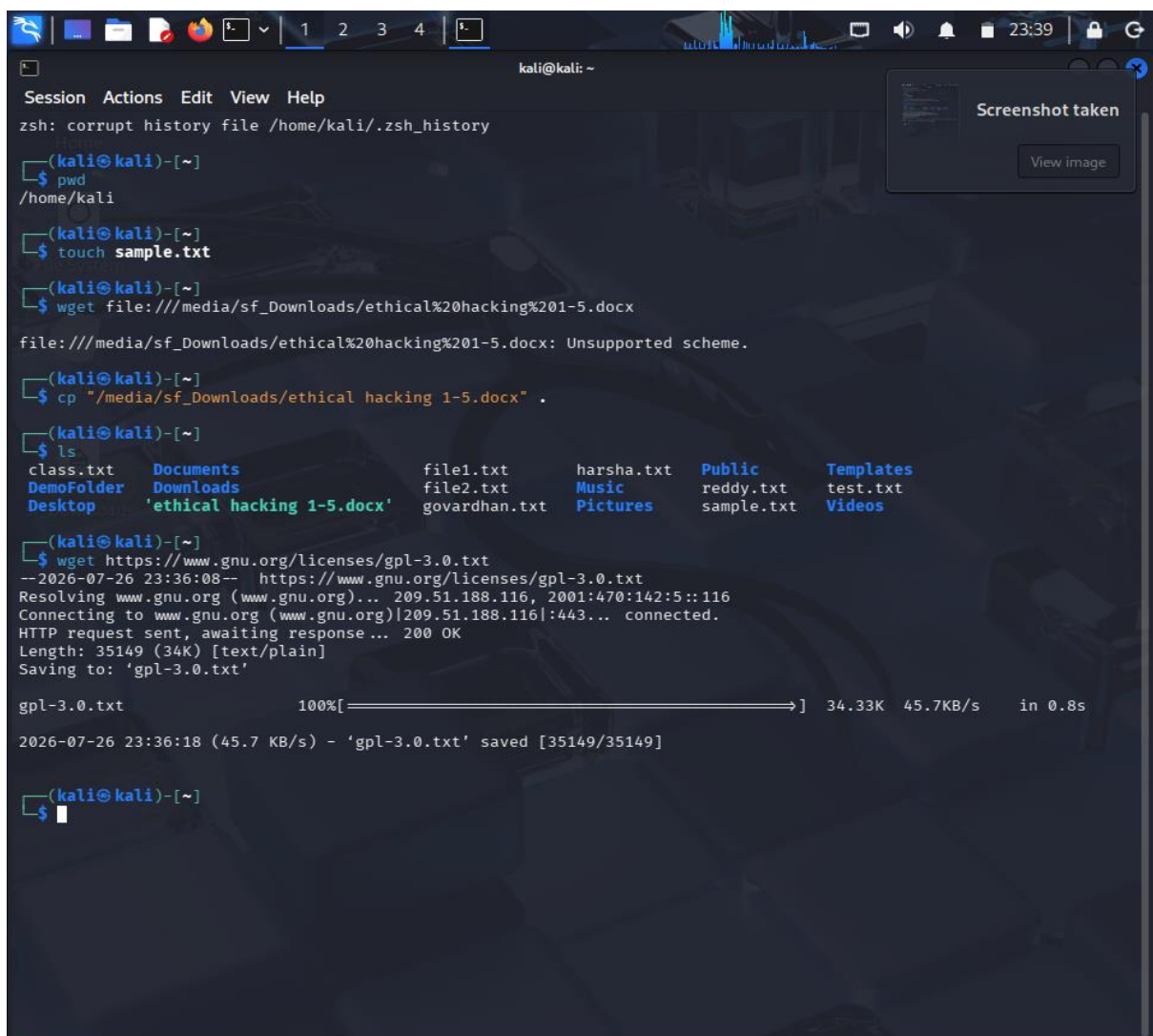
wget Command

Downloads files from the internet.

wget https://example.com/file.zip

Procedure

1. Create a file using touch.
2. Display current directory using pwd.
3. Download a sample file using wget.



```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
  
(kali@kali)-[~]  
$ pwd  
/home/kali  
  
(kali@kali)-[~]  
$ touch sample.txt  
  
(kali@kali)-[~]  
$ wget file:///media/sf_Downloads/ethical%20hacking%201-5.docx  
file:///media/sf_Downloads/ethical%20hacking%201-5.docx: Unsupported scheme.  
  
(kali@kali)-[~]  
$ cp "/media/sf_Downloads/ethical hacking 1-5.docx" .  
  
(kali@kali)-[~]  
$ ls  
class.txt      Documents      file1.txt      harsha.txt     Public         Templates  
DemoFolder    Downloads      file2.txt      Music          reddy.txt      test.txt  
Desktop        'ethical hacking 1-5.docx'  govardhan.txt  Pictures       sample.txt     Videos  
  
(kali@kali)-[~]  
$ wget https://www.gnu.org/licenses/gpl-3.0.txt  
--2026-07-26 23:36:08-- https://www.gnu.org/licenses/gpl-3.0.txt  
Resolving www.gnu.org (www.gnu.org)... 209.51.188.116, 2001:470:142:5::116  
Connecting to www.gnu.org (www.gnu.org)|209.51.188.116|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 35149 (34k) [text/plain]  
Saving to: 'gpl-3.0.txt'  
  
gpl-3.0.txt      100%[=====] 34.33K  45.7KB/s  in 0.8s  
2026-07-26 23:36:18 (45.7 KB/s) - 'gpl-3.0.txt' saved [35149/35149]  
  
(kali@kali)-[~]  
$
```

Result

Successfully executed touch, pwd, and wget commands.

Experiment 7

Navigation and Copy Commands

Aim

To execute cd, cp, and ls commands.

Theory

cd Command

Changes the current directory.

cd Documents

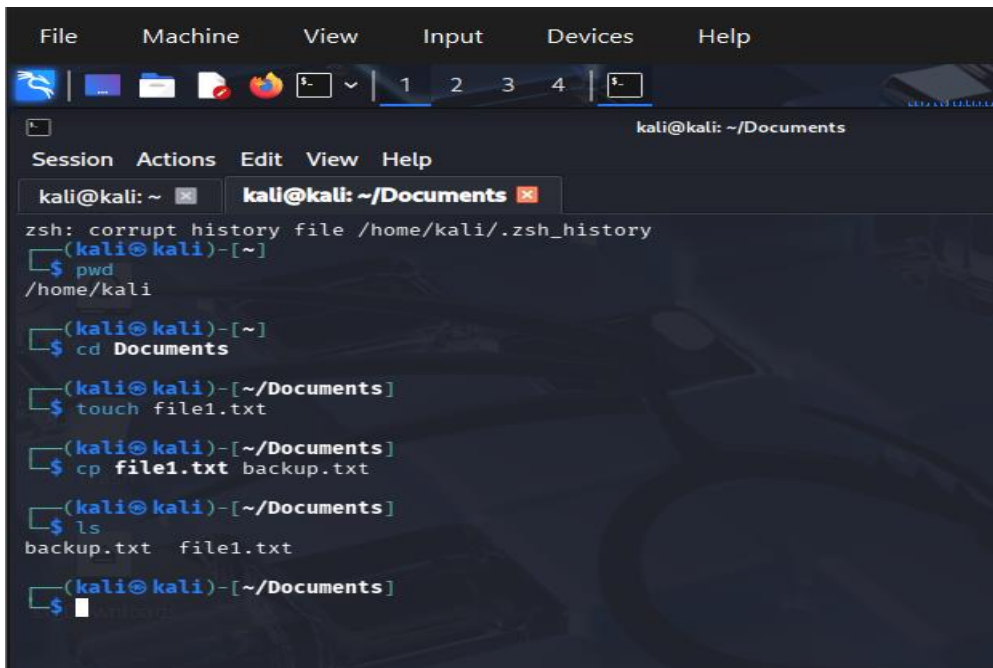
cp Command

Copies files and directories.

cp file1.txt backup.txt

Procedure

1. Navigate between directories using cd.
2. Copy files using cp.
3. Verify copied files using ls.

A screenshot of a Kali Linux terminal window. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu bar is a toolbar with various icons. The terminal title bar shows 'kali@kali: ~/Documents'. The terminal content shows a session with the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ pwd
/home/kali
(kali@kali)-[~]
$ cd Documents
(kali@kali)-[~/Documents]
$ touch file1.txt
(kali@kali)-[~/Documents]
$ cp file1.txt backup.txt
(kali@kali)-[~/Documents]
$ ls
backup.txt  file1.txt
(kali@kali)-[~/Documents]
$
```

Result

Successfully executed cd, cp, and ls commands.

Experiment 8

Nmap Service Version and OS Detection

Aim

To perform service version detection and operating system identification using Nmap.

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands

i) Detect Service Versions

```
nmap -sV 192.168.1.10
```

Description:

Determines the versions of services running on target ports.

ii) Aggressive Scan

```
nmap -A 192.168.1.10
```

Description:

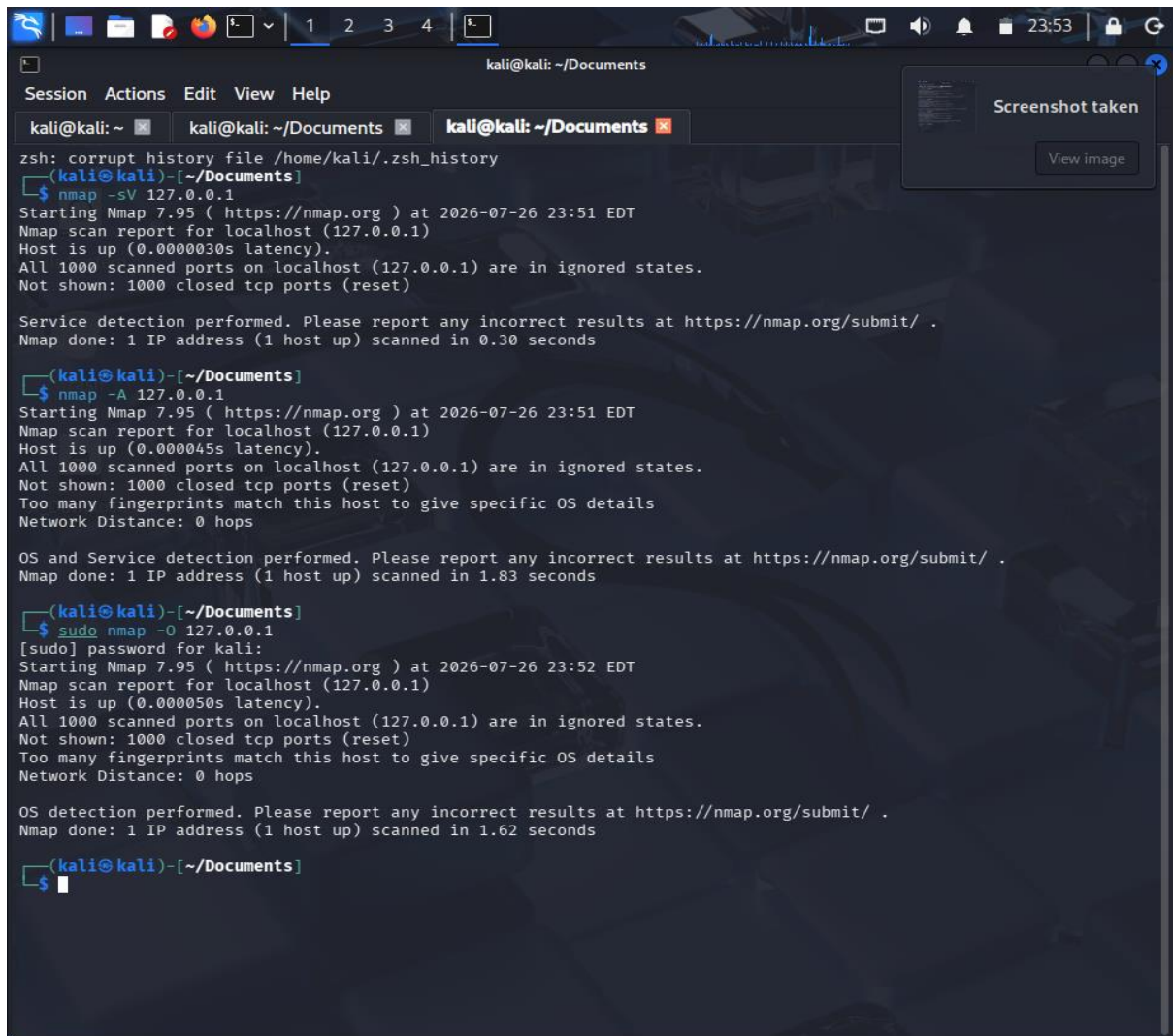
Performs OS detection, version detection, script scanning, and traceroute.

iii) Operating System Detection

```
sudo nmap -O 192.168.1.10
```

Description:

Identifies the operating system running on the target host.



```
kali@kali: ~/Documents
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~/Documents kali@kali: ~/Documents
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Documents]
$ nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:51 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds

(kali@kali)-[~/Documents]
$ nmap -A 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:51 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000045s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.83 seconds

(kali@kali)-[~/Documents]
$ sudo nmap -O 127.0.0.1
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:52 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000050s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds

(kali@kali)-[~/Documents]
$
```

Result

Successfully performed service version detection and OS identification using Nmap.

Experiment 9

Nmap Timing and Performance Commands

Aim

To perform IDS evasion and timing scans using Nmap.

Theory

Timing options control the speed and stealthiness of Nmap scans.

Commands

i) Polite IDS Evasion

`nmap -T2 192.168.1.10`

Description:

Slows down scanning to reduce the chance of detection.

ii) Normal Scan

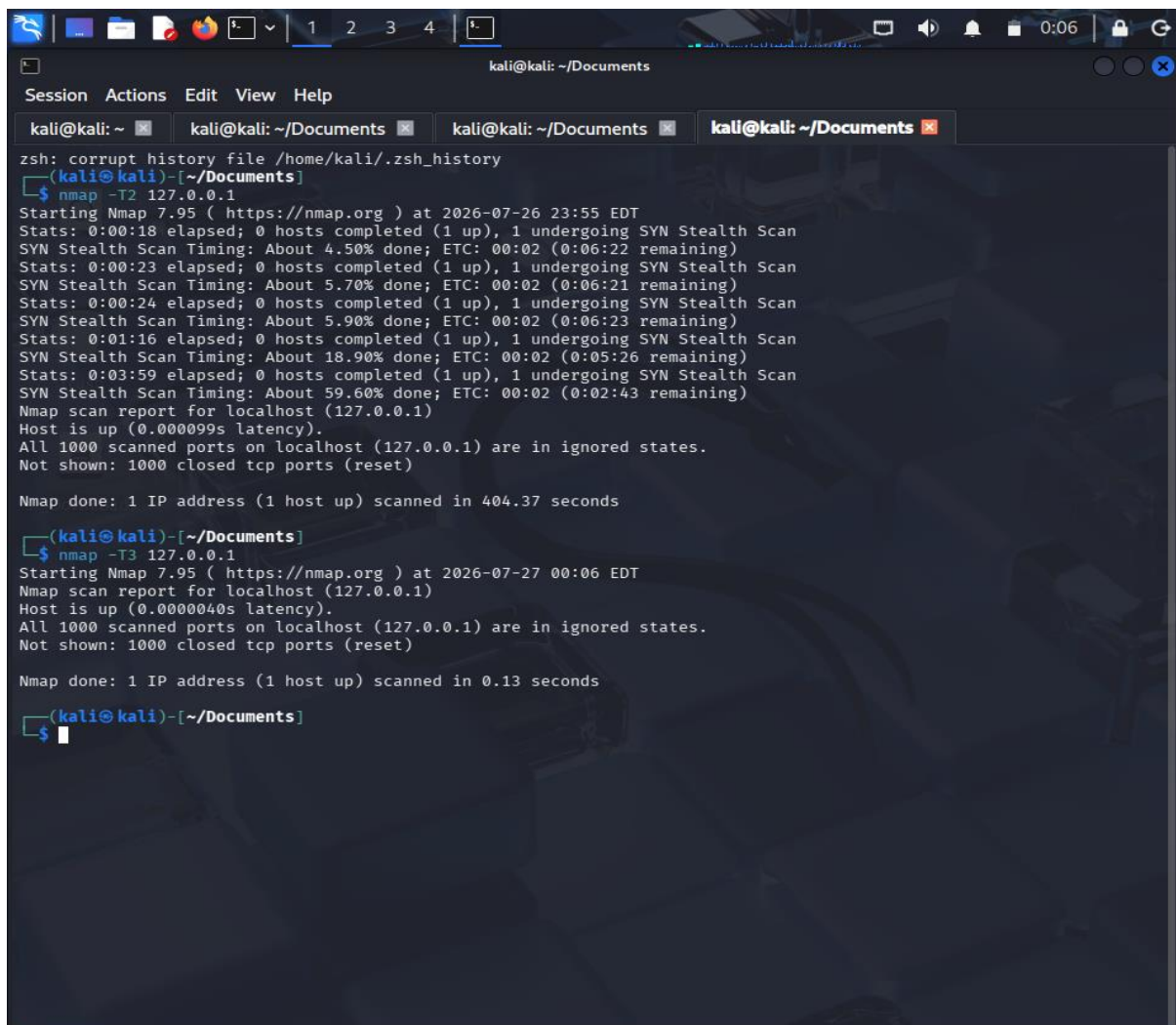
`nmap -T3 192.168.1.10`

Description:

Default balanced scanning speed.

Procedure

1. Identify target IP address.
2. Execute Nmap timing commands.
3. Observe scan duration and results.
4. Compare scan speeds.



```
kali@kali: ~/Documents
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~/Documents kali@kali: ~/Documents kali@kali: ~/Documents
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~/Documents]
$ nmap -T2 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-26 23:55 EDT
Stats: 0:00:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 4.50% done; ETC: 00:02 (0:06:22 remaining)
Stats: 0:00:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.70% done; ETC: 00:02 (0:06:21 remaining)
Stats: 0:00:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.90% done; ETC: 00:02 (0:06:23 remaining)
Stats: 0:01:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 18.90% done; ETC: 00:02 (0:05:26 remaining)
Stats: 0:03:59 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 59.60% done; ETC: 00:02 (0:02:43 remaining)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000099s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 404.37 seconds

(kali@kali)-[~/Documents]
$ nmap -T3 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-27 00:06 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds

(kali@kali)-[~/Documents]
$
```

Result

Successfully executed Nmap timing and performance scans in Kali Linux.

Experiment 10: Free, Sort and History Commands

Aim:

To execute and understand the usage of Free, Sort, and History commands in Kali Linux.

Commands:

free

```
free -h
```

sort file.txt

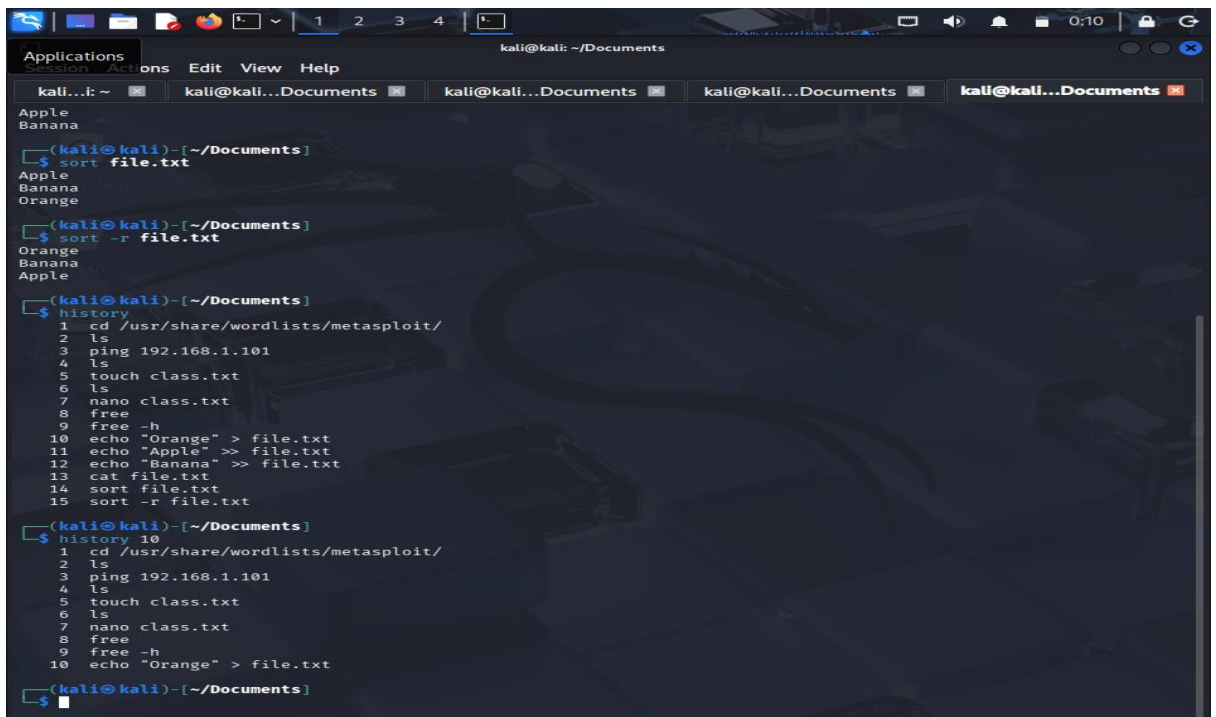
```
sort -r file.txt
```

history

```
history 10
```

Procedure:

1. Open Terminal in Kali Linux.
2. Execute the `free` command to view memory usage.
3. Create a text file and use the `sort` command.
4. Execute the `history` command to display previously executed commands.



```
kali@kali: ~/Documents
Applications Session Actions Edit View Help
kali...: ~ kali@kali...Documents kali@kali...Documents kali@kali...Documents kali@kali...Documents
Apple
Banana
(kali@kali)-[~/Documents]
$ sort file.txt
Apple
Banana
Orange
(kali@kali)-[~/Documents]
$ sort -r file.txt
Orange
Banana
Apple
(kali@kali)-[~/Documents]
$ history
1 cd /usr/share/wordlists/metasploit/
2 ls
3 ping 192.168.1.101
4 ls
5 touch class.txt
6 ls
7 nano class.txt
8 free
9 free -h
10 echo "Orange" > file.txt
11 echo "Apple" >> file.txt
12 echo "Banana" >> file.txt
13 cat file.txt
14 sort file.txt
15 sort -r file.txt
(kali@kali)-[~/Documents]
$ history 10
1 cd /usr/share/wordlists/metasploit/
2 ls
3 ping 192.168.1.101
4 ls
5 touch class.txt
6 ls
7 nano class.txt
8 free
9 free -h
10 echo "Orange" > file.txt
(kali@kali)-[~/Documents]
$
```

Result:

Successfully executed Free, Sort, and History commands in Kali Linux.
